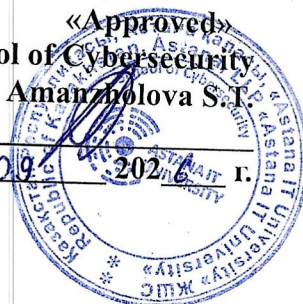




«Approved»
Dean of School of Cybersecurity
Amanzholova S.T.

« 02 » 09 2026 г.

Syllabus
Academic Year 2026 - 2027



1. General information					
Course Code	OS 2207				
Course Title	Operating Systems				
Degree Cycle (Level)/ Major / Relation to curriculum	Bachelor's Degree 6B06301 – Cyber Security БД Bachelor's Degree 6B06202 - Smart Technology БД				
Year, trimester	2 st year, 4				
Language of Instruction:	English				
Lecturer(s)/ Instructors/ Instructor Contact Information	Kymbat Seilkhanova, Senior-Lecturer, MSc in Engineering K.Seilkhanova@astanait.edu.kz Dana Tolegen, Senior-Lecturer, MSc in Engineering D.Tolegen@astanait.edu.kz Laura Kulbayeva, Senior-Lecturer, MSc kulbayeva@astanait.edu.kz Yernar Gabiden, Senior Lecturer, MSc in Technical Science, Yernar.Gabiden@astanait.edu.kz Teili Makasheva, Teacher, MSc in Computer Science Email: Teili.Makasheva@astanait.edu.kz Mukusheva Sania, MSc in Computer Science S.Mukusheva@astanait.edu.kz Togissova Akerke, MSc in Engineering, Teacher A.Togissova@astanait.edu.kz Maksat Meshituly, senior lecturer, MSc in Computer science m.meshituly@astanait.edu.kz Meiirgali Mussylmanbay, Senior-Lecturer, MSc in Data Science m.mussylmanbay@astanait.edu.kz Kuanysh Syzdykbayev, Senior-Lecturer, MSc in Technical Science K.Syzdykbayev@astanait.edu.kz Turysbek Nazym, Senior-Lecturer, MSc Advanced Computing, Turysbek.nazym@gmail.com Kali Marzhan, Senior-Lecturer, MSc in Technical Science marzhan.kali@astanait.edu.kz Turysbek Nazym, Senior-Lecturer, MSc Advanced Computing, Turysbek.nazym@gmail.com Contact details: Expo C1, Room C1.1.329				
Number of Credits	5				
Workload of course components and credits per trimester					
	ECTS credits	Contact hours		IASS	SIS Total hours

	5. Oracle® VM VirtualBox® 6. Best Online Linux Terminals and Online Bash Editors (itsfoss.com)	
8. Technology employed:	The course employs a blended technological environment that combines Moodle (moodle.astanait.edu.kz) as the primary Learning Management System for announcements, assignment submissions, quizzes, and exams; Microsoft Teams for synchronous online sessions, consultations, and webinars when in-person instruction is not possible. Laboratory sessions will be conducted using virtual machines based on Oracle VirtualBox, which will allow learners to safely experiment with different environments. Both Linux and Windows operating systems will be used to study system configuration, command-line operations, and application of theoretical concepts. This combination ensures that students gain hands-on skills in managing and analyzing multiple platforms. In addition, students will use Cisco Networking Academy resources, online journals, and professional articles to support independent study and enhance theoretical knowledge. In accordance with institutional teaching standards, students are expected to have consistent access to internet-enabled devices to engage effectively with LMS, communication platforms, and simulation tools.	
9. <u>Course Policies</u>	<u>Course Policies for In-Person Instruction:</u> - <u>Assignments Policy</u> The course includes a structured List of Assignments, outlining weekly topics and tasks. Students will be assigned laboratory work at one of three complexity levels: Easy, Medium, or Hard. Each level is associated with a distinct point value, reflecting its degree of difficulty. - <u>Presence and Participation</u> Students are expected to attend all scheduled class sessions with the required readings and supplementary materials completed beforehand. Attendance itself does not provide extra points, but it is mandatory for course completion. Students must participate in at least 70% of all class sessions; otherwise, they will not be admitted to the exam and will fail the course. Active participation is expected and will be evaluated by the quality of contributions, such as offering unique perspectives, advancing discussion, and building on peers' comments. Being more than 15 minutes late results in being marked absent. - <u>Deadlines and Late Submissions</u> All assignments and homework must be submitted on time. Since assignments are often discussed in class on the due date, late submissions will not be accepted. Submissions through Moodle after the deadline will automatically reduce the overall score by 30%, while late defenses of assignments result in 0%. In exceptional circumstances, students may request an extension in advance with proper justification; if approved, a new deadline will be assigned. - <u>The Honor Code of Students</u> Cheating, plagiarism, and other forms of academic dishonesty are strictly prohibited. Examples include submitting work that is not your own, copying from sources without citation, falsifying data, or submitting the same work in two courses without consent. Any violation will result in a grade of 0 for the assignment and may lead to additional disciplinary measures under the University's Academic Conduct Policies. - <u>Course Policies for Use of Generative AI</u> Generative AI tools (e.g., ChatGPT, Copilot) may only be used when explicitly allowed by the instructor. If permitted, students must clearly acknowledge the use of such tools and provide proper citations where relevant. Using AI to generate uncited content or to complete assignments dishonestly is considered plagiarism and will be subject to disciplinary measures. - <u>Extended Absences</u> Students unable to attend class for extended periods must inform the instructor in advance and are responsible for covering missed material independently. Extended absences without a valid reason may prevent successful course completion. If more than 30% of classes are missed without excuse, the student will be marked "not graded" and denied access to the final exam. - <u>Late Homework Policy</u> Late homework is generally not accepted. Work submitted after the deadline in Moodle will receive a 30% point deduction, and late defenses earn 0%. Exceptions are only made for extraordinary events, with prior approval and evidence, in which case a new deadline may be assigned. - <u>Exam Policy</u>	

	Two mid-term and end-term exams, along with a final exam, will be held. Exams include both theoretical (multiple-choice) and practical parts. Attendance and participation requirements must be met to qualify for exams. Use of unauthorized materials, electronic devices, or cheating during exams is strictly prohibited and will result in disciplinary action. - <u>Course Privacy Statement</u> Course materials, lecture slides, assignments, and exams are provided for personal academic use only. Recording or sharing course content without instructor approval is prohibited. Student contributions in class are treated as part of a private academic setting and should not be disclosed outside the course environment without consent.
10. <u>Course Schedule</u>	The course Operating Systems is designed as a 10-week program consisting of lectures, practice sessions, teacher-supervised independent study (TSIS), and student independent study (SIS), with a total workload of 150 hours (5 ECTS). Each week covers both theoretical concepts and practical applications, supported by independent assignments. Topics progress from the fundamentals of operating systems to advanced issues of system management and performance. Week 1 – Introduction to OS. Basic Concepts in Operating Systems. System Architecture. Week 2 – Operating systems structure. The Shell. Command Line Skills. Week 3 – Building and Booting Operating systems. Navigation. Working with Files and Directories. Week 4 – Communication between threads and processes. Filter Contents. Regular Expressions. Permission Management Week 5 – Workflow. Service management Week 6 – Virtual memory System Management. Basic Scripting. Week 7 – I/O Management and Disk Scheduling Network Services. Week 8 – File system System and User Security. Week 9 – Operating system security. Linux Administration. Week 10 – Implementation of network capabilities. Topics review Each week includes 2 lecture hours, 3 practice hours, 1 TSIS hour, and 9 SIS hours, with independent assignments focusing on lab-based tasks such as working with Linux and Windows operating systems, virtual machines, shell scripting, system security, file permissions, and backup management. Continuous assessment is conducted through weekly assignments, mid-term and end-term evaluations, followed by a final exam.
11. <u>Syllabus Inclusion Statements</u>	The course provides conditions for inclusive education and offers students with special educational needs (SEN) additional opportunities, including the adaptation of teaching methods, formats, and tools: • individualized consultations and clarification of learning materials; • the use of multimedia and assistive technologies (screen readers, subtitles, enlarged font, etc.); • provision of alternative formats for mastering the material (audio materials, visual diagrams, step-by-step instructions); • implementation of flexible participation formats in the learning process (online attendance, remote submission of certain assignments, use of written instead of oral formats, etc.). Students with SEN are provided with: • additional time for completing written, test, and project assignments within the limits established by the academic policy; • adapted assignments that maintain academic standards while taking individual needs into account (e.g., simplifying the format of presenting results without reducing the content requirements); • the possibility to choose assessment formats provided by the academic policy (oral/written,

	online/offline), if required due to SEN. A student with SEN has the right to inform the teacher about the need for adaptation at the beginning of the trimester or at any time when such a need arises. The teacher ensures the confidentiality of information about the student's health and individual characteristics. If necessary, an individual support plan is developed in coordination with the Registrar's Office, within which assessment activities, assignment deadlines, and forms of interaction may be adapted. The course promotes respect for diverse ideas and experiences, encourages critical engagement with different viewpoints, and emphasizes academic integrity and professional conduct. Students are also expected to respect the privacy of class discussions and materials, and to treat their peers and instructors with courtesy and professionalism at all times.	
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3.1 Abbreviations

#	Abbreviation	Meaning
1	IASS	Instructor assisted student study - Instructors assist students in achieving success by delivering support, feedback, and resources. This support might take the form of clarifying difficult topics, recommending additional materials, or providing individualized guidance.
2	SIS	Student independent study
3	IW	Individual work
4	GW	Group work
5	PA	Practical assignment
6	LW	Laboratory work
7	MCQ	Multiple choice quiz
8	PBT	Paper-based test

NB: You may introduce abbreviations that are used in the syllabus

3.2 Course Plan

Week No	Course Topic	Lectures (h/w)	Practice sessions (h/w)	Lab. Sessions (h/w)	IASS (h/w)	SIS (h/w)
1	Introduction to OS. Basic Concepts in Operating Systems. System Architecture.	2	3	0	1	9
2	Operating systems structure. The Shell. Command Line Skills.	2	3	0	1	9
3	Building and Booting Operating systems. Navigation. Working with Files and Directories.	2	3	0	1	9
4	Communication between threads and processes. Filter Contents. Regular Expressions. Permission Management	2	3	0	1	9
5	Workflow. Service management	2	3	0	1	9

6	Virtual memory System Management. Basic Scripting.	2	3	0	1	9
7	I/O Management and Disk Scheduling Network Services.	2	3	0	1	9
8	File system System and User Security.	2	3	0	1	9
9	Operating system security. Linux Administration.	2	3	0	1	9
10	Implementation of network capabilities. Topics review	2	3	0	1	9
	Total: 150 hours	20	20	0	10	90

3.3 Detailed course plan

Week No	Lecture Topics with Learning Resources	Tasks and Activities for Practice Sessions with Recommended Reading Resources	Lab Exercises, Assigned Activities, and Supporting Reading Resources (Some courses do not include lab work; in such cases, this column can be omitted)
1	Week 1: Introduction to OS. Basic Concepts in Operating Systems. System Architecture. - <i>Modern Operating Systems (Tanenbaum & Bos, 2022)</i>	Introduction to Linux Operating Systems Working in Linux - <i>Linux Essentials (NDG, 2023)</i>	-
2	Week 2: Operating systems structure. The Shell. Command Line Skills. - <i>Modern Operating Systems (Tanenbaum & Bos, 2022)</i>	Operating system models: monolithic, multi-tier, virtual machines Open-Source Software and Licensing Command Line Skills Getting Help - <i>Online Linux Terminals (ItsFOSS)</i> - <i>Oracle VM VirtualBox (Oracle)</i>	-
3	Week 3: Building and Booting Operating systems. Navigation. Working with Files and Directories. - <i>Modern Operating Systems (Tanenbaum & Bos, 2022)</i>	Classification of processes in the operating system Navigating the Filesystem Managing Files and Directories Archiving and Compression - <i>Linux Essentials (NDG, 2023)</i> - <i>Online Linux Terminals (ItsFOSS)</i>	-
4	Week 4: Communication between threads and processes. Filter Contents. Regular Expressions. Permission Management - <i>Understanding Operating Systems (Ann & Ida, 2019)</i>	Working with BIOS Filter Contents. Regular Expressions. Permission Management - <i>Linux Essentials (NDG, 2023)</i>	-
5	Week 5: Workflow. Service management - <i>Modern Operating Systems (Tanenbaum & Bos, 2022)</i>	Computer memory hierarchy - <i>Linux Essentials (NDG, 2023)</i> - <i>Ubuntu Documentation (Canonical)</i>	-

6	Week 6: Virtual memory System Management. Basic Scripting - <i>Modern Operating Systems (Tanenbaum & Bos, 2022)</i>	Paging and paging management Working with Text Basic Scripting Understanding - <i>Linux Essentials (NDG, 2023)</i> - <i>Ubuntu Documentation (Canonical)</i>	-
7	Week 7: I/O Management and Disk Scheduling Network Services. - <i>Modern Operating Systems (Tanenbaum & Bos, 2022)</i> - <i>Operating System Concepts (Silberschatz, Galvin & Gagne, 2018)</i>	Mechanisms for managing the power consumption of external devices Where Data is Stored Network Configuration - <i>Linux Essentials (NDG, 2023)</i> - <i>Ubuntu Documentation (Canonical)</i>	-
8	Week 8: File system System and User Security. - <i>Modern Operating Systems (Tanenbaum & Bos, 2022)</i>	Android File System System and User Security Creating Users and Groups - <i>Linux Essentials (NDG, 2023)</i> - <i>Ubuntu Documentation (Canonical)</i>	-
9	Week 9: Operating system security. Linux Administration. - <i>Modern Operating Systems (Tanenbaum & Bos, 2022)</i> - <i>Operating Systems (Arpaci-Dusseau & Arpaci-Dusseau, 2018)</i>	Access rights distribution models Linux Administration. Ownership and Permissions Special Directories and Files - <i>Linux Essentials (NDG, 2023)</i> - <i>Ubuntu Documentation (Canonical)</i>	-
10	Week 10: Implementation of network capabilities. Topics review - <i>Understanding Operating Systems (Ann & Ida, 2019)</i>	Diagnosing network connections - <i>Linux Essentials (NDG, 2023)</i> - <i>Ubuntu Documentation (Canonical)</i>	-

3.4 List of assignments for Student Independent Study

№	Assignments (topics) for Independent Study	Hours	Recommended literature and other sources (links)	Form of submission and deadlines
1	2	3	4	5
1	Introduction to Linux Operating Systems Working in Linux	9	(<i>Linux Essentials NDG, 2023;</i> <i>Modern Operating Systems, Tanenbaum & Bos, 2022)</i>	Labs
2	Operating system models: monolithic, multi-tier, virtual machines Open-Source Software and Licensing Command Line Skills Getting Help	9	(<i>Linux Essentials NDG, 2023;</i> <i>Modern Operating Systems, Tanenbaum & Bos, 2022)</i>	Labs
3	Classification of processes in the operating system Navigating the Filesystem Managing Files and Directories Archiving and Compression	9	(<i>Linux Essentials NDG, 2023;</i> <i>Modern Operating Systems, Tanenbaum & Bos, 2022)</i>	Labs

4	Working with BIOS Filter Contents. Regular Expressions. Permission Management	9	(Linux Essentials NDG, 2023; Modern Operating Systems, Tanenbaum & Bos, 2022)	Labs
5	Computer memory hierarchy	9	(Linux Essentials NDG, 2023; Modern Operating Systems, Tanenbaum & Bos, 2022)	Labs
6	Paging and paging management Working with Text Basic Scripting Understanding	9	(Linux Essentials NDG, 2023; Modern Operating Systems, Tanenbaum & Bos, 2022)	Labs
7	Mechanisms for managing the power consumption of external devices Where Data is Stored Network Configuration	9	(Linux Essentials NDG, 2023; Modern Operating Systems, Tanenbaum & Bos, 2022)	Labs
8	Android File System System and User Security Creating Users and Groups	9	(Linux Essentials NDG, 2023; Modern Operating Systems, Tanenbaum & Bos, 2022)	Labs
9	Access rights distribution models Linux Administration. Ownership and Permissions Special Directories and Files	9	(Linux Essentials NDG, 2023; Modern Operating Systems, Tanenbaum & Bos, 2022)	Labs
10	Diagnosing network connections	9	(Linux Essentials NDG, 2023; Modern Operating Systems, Tanenbaum & Bos, 2022)	Labs

3.5 For instructor assisted student study (IASS) schedule and policy follow your course LMS Moodle.

4. Student performance evaluation system for the course.

Period	Assignments*	Number of points	Form of assessment	Total
1 st attestation	Assignment 1 (week 1)	20		100
	Assignment 2 (week 2)	20		
	Assignment 3 (week 3)	20		
	Assignment 4 (week 4)	20		
	Midterm exam	20	MCQ	
2 nd attestation	Assignment 5 (week 6)	20		100
	Assignment 6 (week 7)	20		
	Assignment 7 (week 8)	20		
	Assignment 8 (week 9)	20		
	Endterm exam	20	MCQ	
Final exam	Exam (Paper based MCQ + Open Questions)			100
Total	0.3*1st Att + 0.3 *2nd Att + 0.4*Final			100

- * The complete title of the assignment must be clearly stated.
- ** The number of assignments may vary depending on the course content and is determined by the course syllabus.
- *** Description of iROS or Social GPA elements, if applicable

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	GPA	Percentage	Description
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	
B	3,0	80-84	Good
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	
C-	1,67	60-64	Satisfactory
D+	1,33	55-59	
D	1,0	50-54	
FX	0	30-49	
F	0	0-29	Fail

Course Assessment Rubric for Student Performance Evaluation

Learning Outcomes	Indicators	Excellent (90–100)	Good (70–89)	Satisfactory (50–69)	Fail (0–49)
1. Introduction to Linux & OS Fundamentals (Introduction to Linux, Operating Systems, OS Models, Open-Source Licensing)	Demonstrates understanding of OS principles, models, licensing	Clearly defines OS concepts and models; compares monolithic, layered, VM structures; explains licensing with examples	Explains main OS concepts and models with minor gaps; shows basic understanding of licensing	Defines OS models and licensing with limited details; lacks depth	Fails to explain OS concepts, models, or licensing
2. Linux Command Line & File Management (Command Line Skills, Getting Help, Navigating Filesystem, Managing Files & Directories, Archiving & Compression)	Applies CLI operations and file management techniques	Executes commands accurately; manages directories, archives, and compression without errors	Performs most CLI and file tasks correctly; occasional errors in syntax or process	Can execute simple commands but struggles with more advanced operations	Unable to perform basic CLI or file management tasks

3.Processes, Memory & BIOS (Classification of Processes, BIOS, Memory Hierarchy, Paging)	Explains and applies process management, memory, BIOS concepts	Accurately classifies processes; explains BIOS and memory hierarchy; demonstrates paging management	Correctly explains most process and memory concepts with small inaccuracies	Understands basic concepts but explanations are unclear or incomplete	Cannot explain or apply process, memory, or BIOS topics
4.Permissions, Security & Administration (Permissions, Regular Expressions, Users/Groups, Access Rights, Ownership, Security Models, Linux Administration)	Configures users, groups, permissions, and applies security models	Correctly manages permissions, users, and groups; applies regex effectively; explains and implements security models	Manages most user and permission settings with small mistakes; partial regex understanding	Performs basic permission or user tasks; struggles with advanced security settings	Fails to manage users, permissions, or apply security models
5. Scripting & Automation (Working with Text, Basic Scripting, Special Directories & Files)	Writes and applies simple shell scripts for automation	Creates well-structured scripts; handles text and files efficiently; demonstrates understanding of automation	Writes functional scripts with minor logic or syntax issues	Writes very simple scripts with limited functionality	Unable to create working scripts
6. Networking & Devices (Network Configuration, Diagnosing Connections, Power Management, Data Storage, Android File System)	Configures networks, storage, and external devices	Accurately configures networks; diagnoses issues; manages storage and power; explains Android FS	Performs most configurations with minor mistakes; explains concepts adequately	Can perform basic configuration but struggles with troubleshooting or advanced storage	Cannot configure or explain networking, storage, or device management

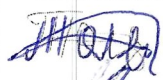
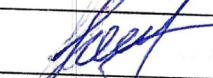
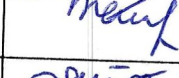
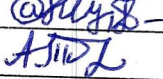
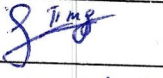
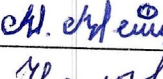
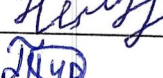
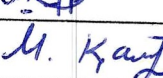
5. Course assessment methodology

Assessment is conducted continuously throughout the course, comprising 60% continuous assessment throughout the semester and 40% summative assessment during the examination session, totaling 100%. Continuous assessment evaluates students' ongoing performance through attendance, laboratory work defense, in-class activities, course assignments as well as independent study (SIS).

- **IASS (Instructor assisted student study)** - Instructors assist students in achieving success by delivering support, feedback, and resources. This support might take the form of clarifying difficult topics, recommending additional materials, or providing individualized guidance.

- **SIS** – Student engagement in self-directed study on assigned topics and the completion of relevant assignments given in the syllabus.
- **Mid-term and End-term** are a review of the topics covered and an assessment of each student's knowledge. The form of the midterm and endterm exams is complex.
- Final assessment for the course "*Operating Systems*" includes a **comprehensive paper-based final exam**, which consists of two parts: (1) a multiple-choice theoretical test covering the fundamental concepts, models, and principles studied during the course, and (2) a practical problem-solving section where students are expected to apply configuration and troubleshooting skills in simulated network scenarios. According to the course exam policy, students must attend at least **70% of the classes** to be admitted to the final assessment. The use of unauthorized materials, communication devices, or any form of cheating during the exam is strictly prohibited and will result in a grade of **0 (zero)**, with further disciplinary actions possible in line with the University Academic Integrity Policy. The final exam accounts for **40% of the overall course grade**, while continuous assessment throughout the semester contributes the remaining 60%.

6. Lecturer (lecturers) signature

Full name	Job title	Date	Signature
Dana Tolegen	Senior-Lecturer, MSc in Engineering	02.09.2026	
Kymbat Seilkhanova	Senior-Lecturer, MSc in Engineering	02.09.2026	
Laura Kulbayeva	Senior-Lecturer, MSc	02.09.2026	
Yernar Gabiden	Senior Lecturer, MSc in Technical Science,	02.09.2026	
Teili Makasheva	Teacher, MSc in Computer Science	02.09.2026	
Mukusheva Sania	MSc in Computer Science	02.09.2026	
Togissova Akerke	MSc in Engineering, Teacher	02.09.2026	
Maksat Meshituly	Senior lecturer, MSc in Computer science	02.09.2026	
Meiirgali Mussylmanbay	Senior-Lecturer, MSc in Data Science	02.09.2026	
Kuanysh Syzdykbayev	Senior-Lecturer, MSc in Technical Science	02.09.2026	
Turysbek Nazym	Senior-Lecturer, MSc Advanced Computing	02.09.2026	
Kali Marzhan	Senior-Lecturer, MSc in Technical Science	02.09.2026	